Lecture 8 - Decision Tree

ECE 447: Data Analysis and Machine Learning for Engineers

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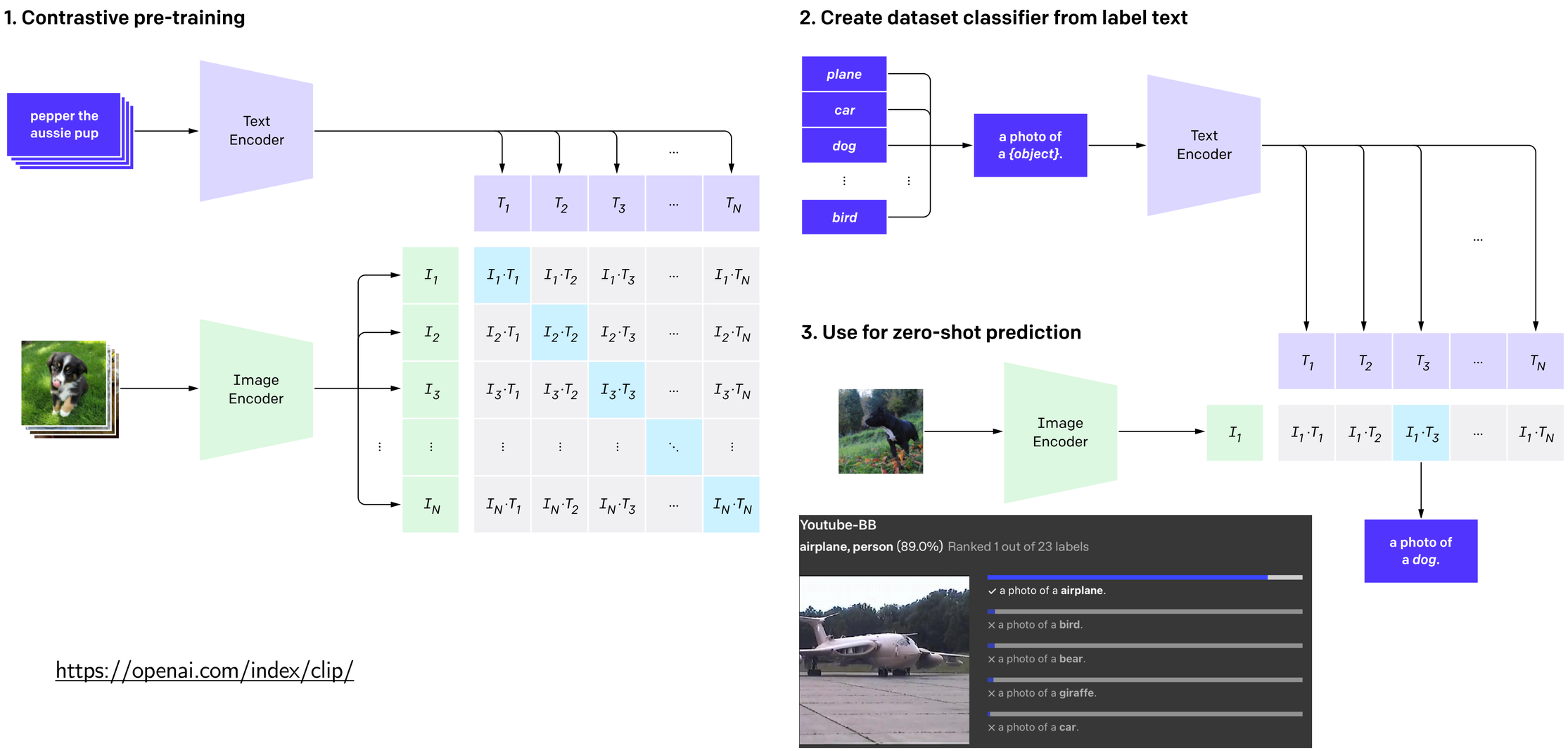
CLIP: Connecting Text and Images

This slide is informational, excluded from assessment.

Contrastive Language-Image Pre-training



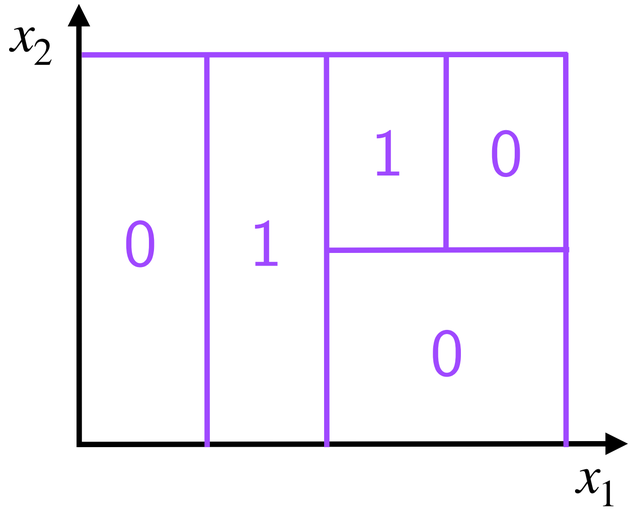
https://openai.com/index/clip/

This Lecture Covers ...

- Decision Tree
  - Entropy and information gain

Intro to Decision Tree Classification

Consider the two-dimensional feature space,



the labels (0,1) correspond to one of the two classes.

Given data $\{(y_i, \boldsymbol{x}_i); i = 1, \ldots, n\}$, a **decision tree** is a classifier that **sequentially** partition the space into distinct rectangles.

Decision Tree Logic of Classification

Decision tree partition the space of feature by

1. Selecting **one dimension** along which to split the data at every iteration.
2. **Recursively** performing such splits.

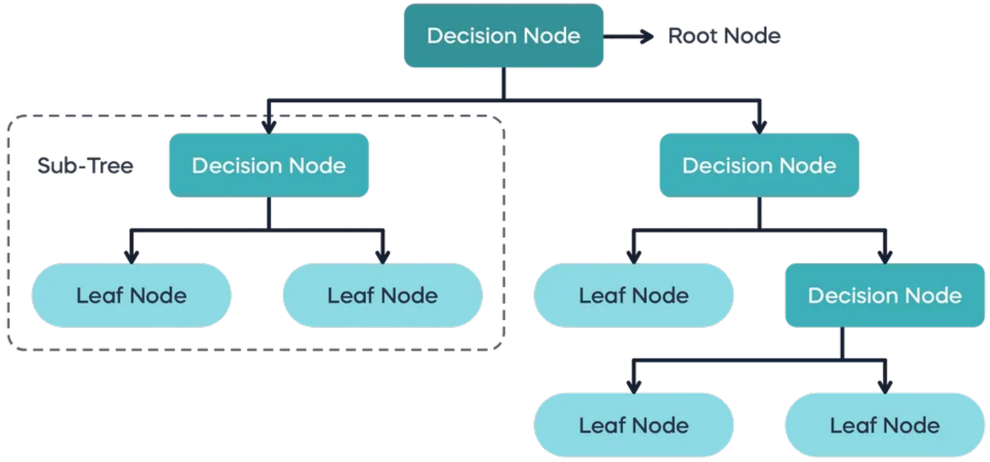
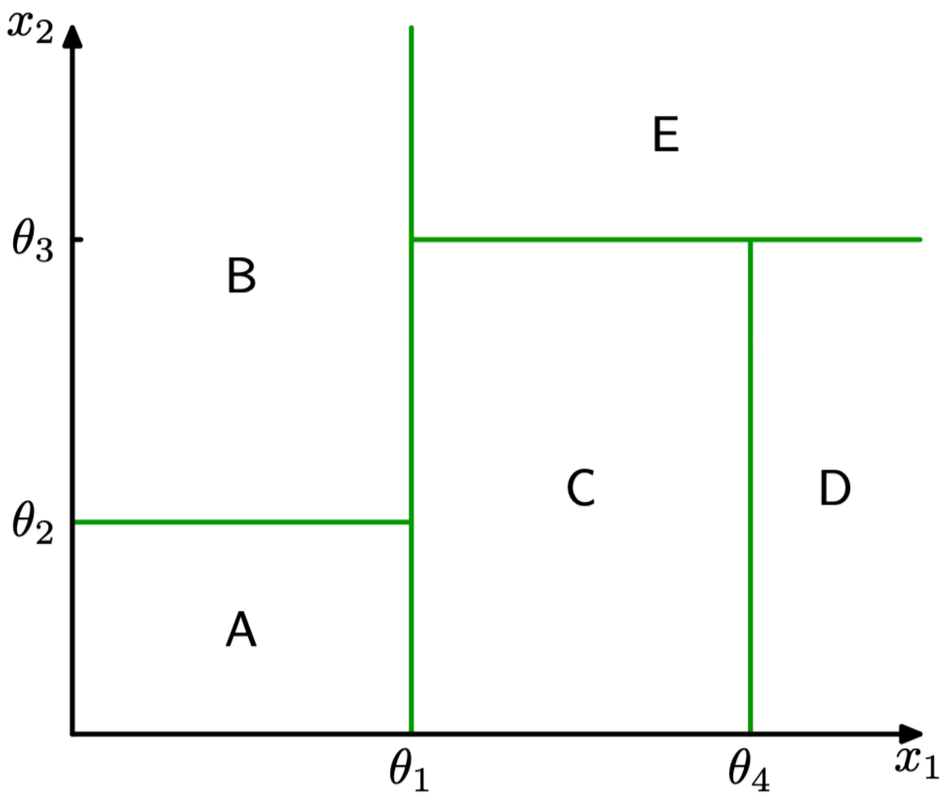
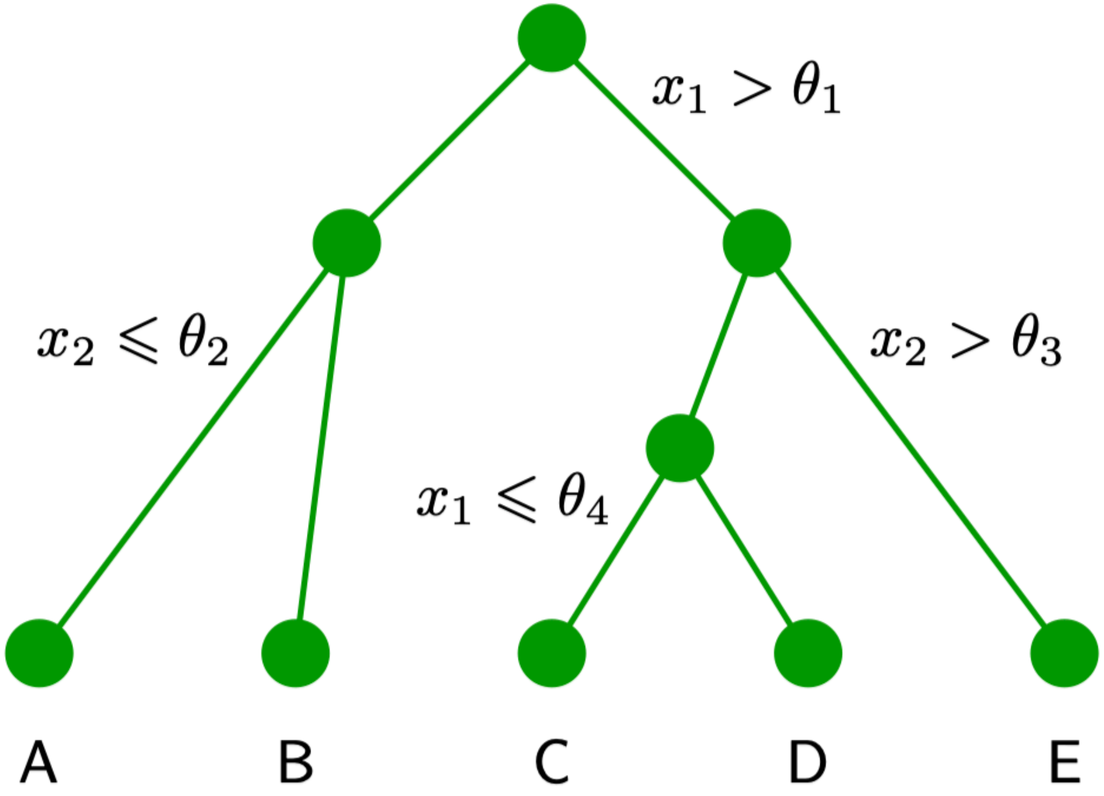


figure from https://365datascience.com/tutorials/machine-learning-tutorials/decision-trees/

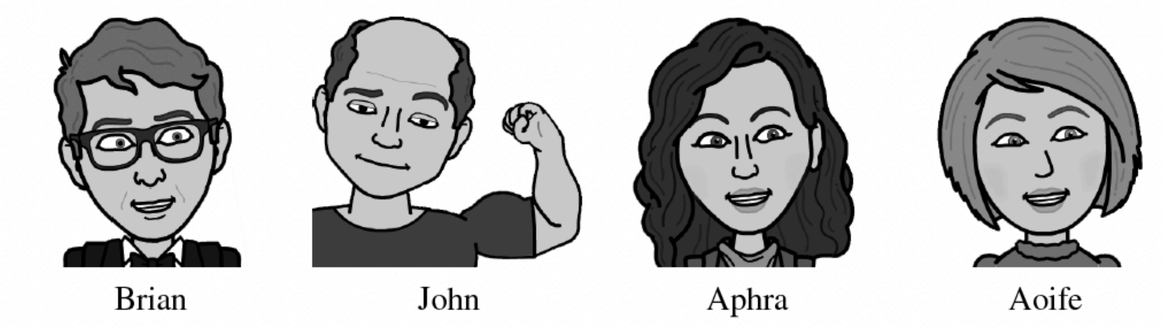
Partition Example



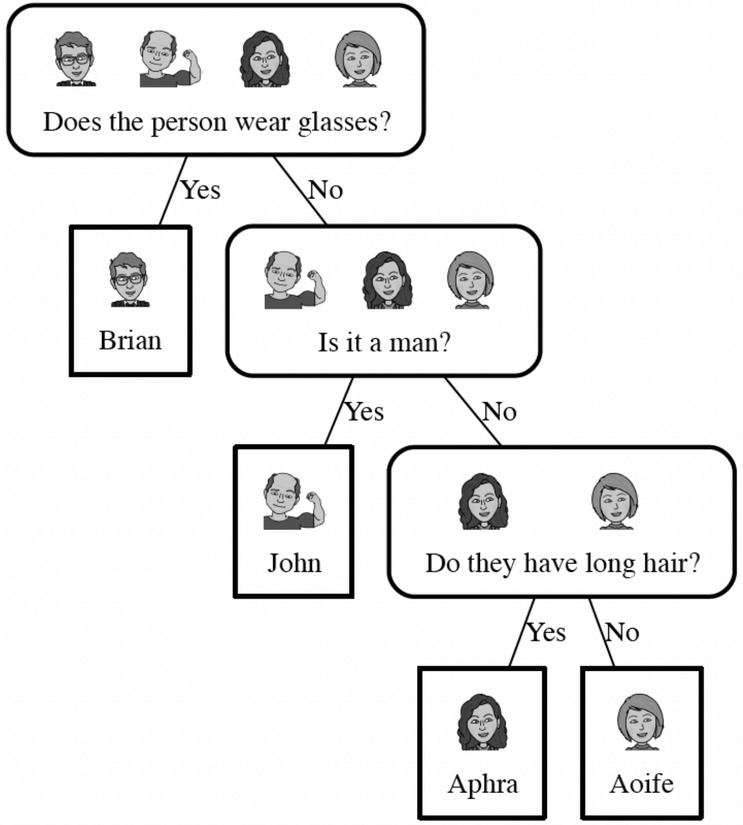


How Does the Partitioning Criteria Work?

Lets play *Guess Who!*



1. Is it a man?
2. Does the person wear glasses?

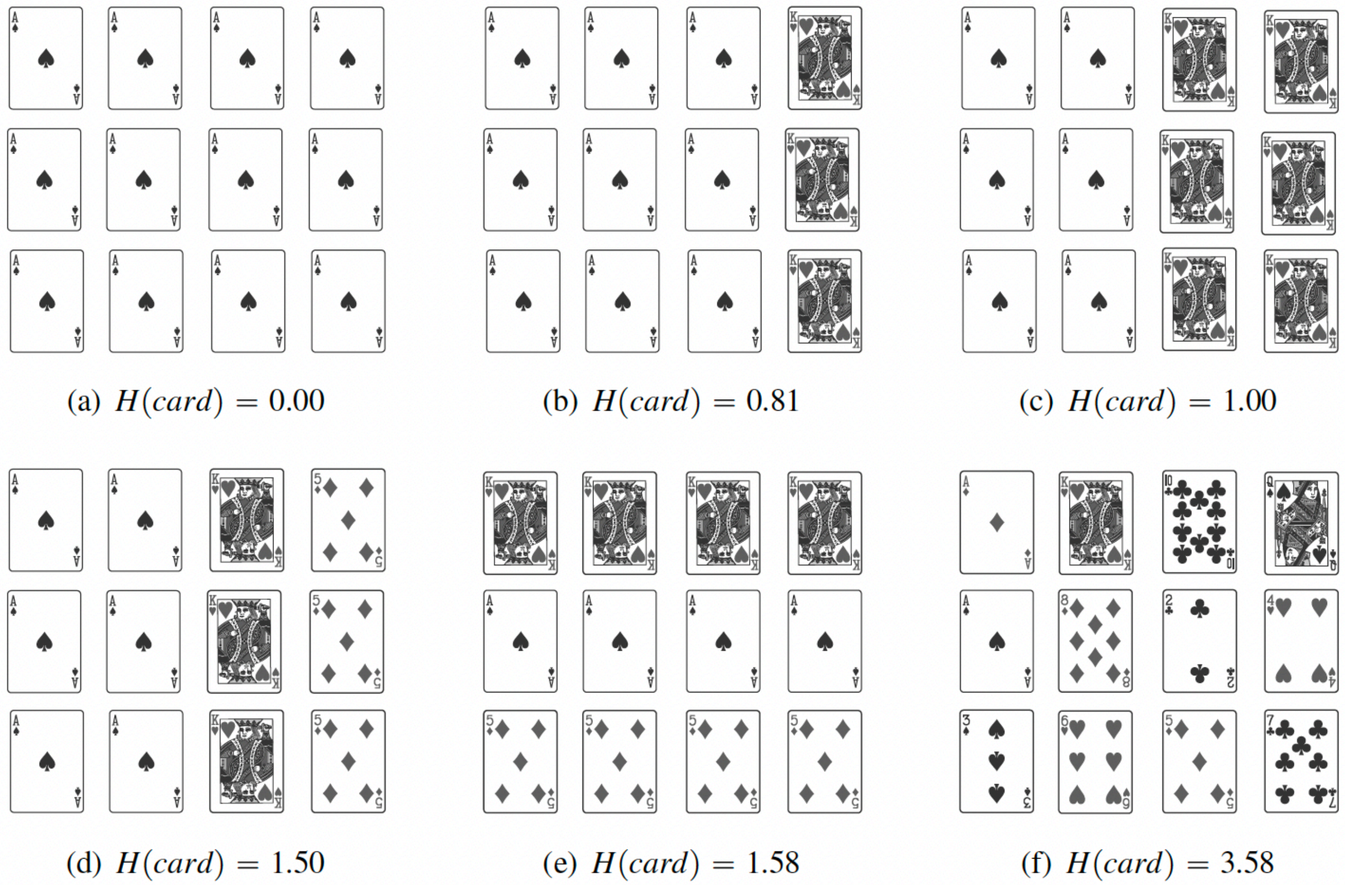


You want to ask questions (i.e., have the decision criteria) that reduce the level of ***uncertainty*** *or give you more* ***information****.*

How to Define the Partitioning Criteria

We use two concepts from information theory:

entropy
and
information gain.



(a) $H(card) = 0.00$ (b) $H(card) = 0.81$ (c) $H(card) = 1.00$

(d) $H(card) = 1.50$ (e) $H(card) = 1.58$ (f) $H(card) = 3.58$

Measure the uncertainty associated with guessing the results if you were to make a random selection from the set.

Claude Shannon's Entropy Model

A computational measure
of the uncertainty.

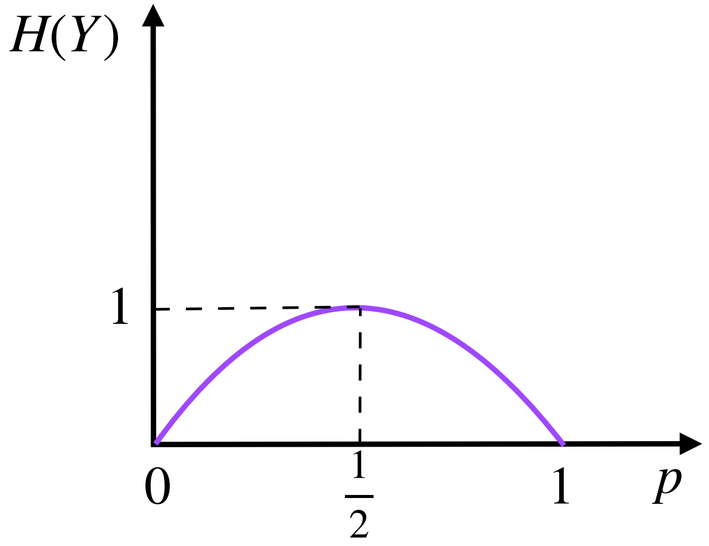
An outcome with
a large probability -> low entropy value,
a small probability -> a large entropy

Let $Y \in \{0,1\}$ be a binary random variable such that $p(Y = y_i) = p_i$. The entropy of Y is

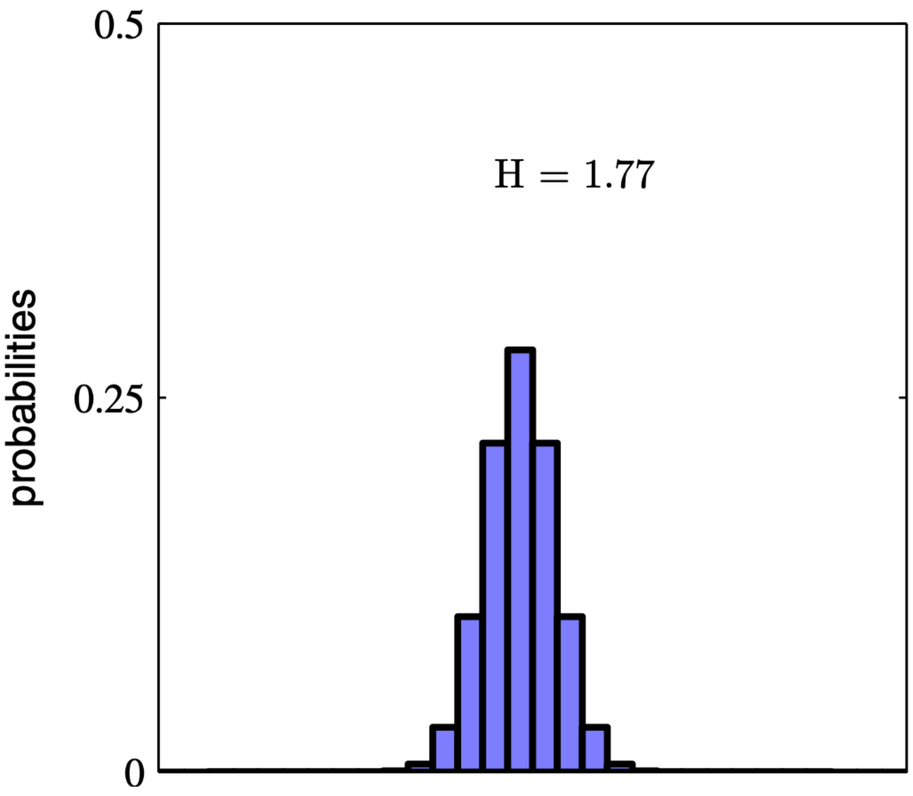
$$H(Y) = \mathbb{E}[-\log_2 p(Y = k)]$$

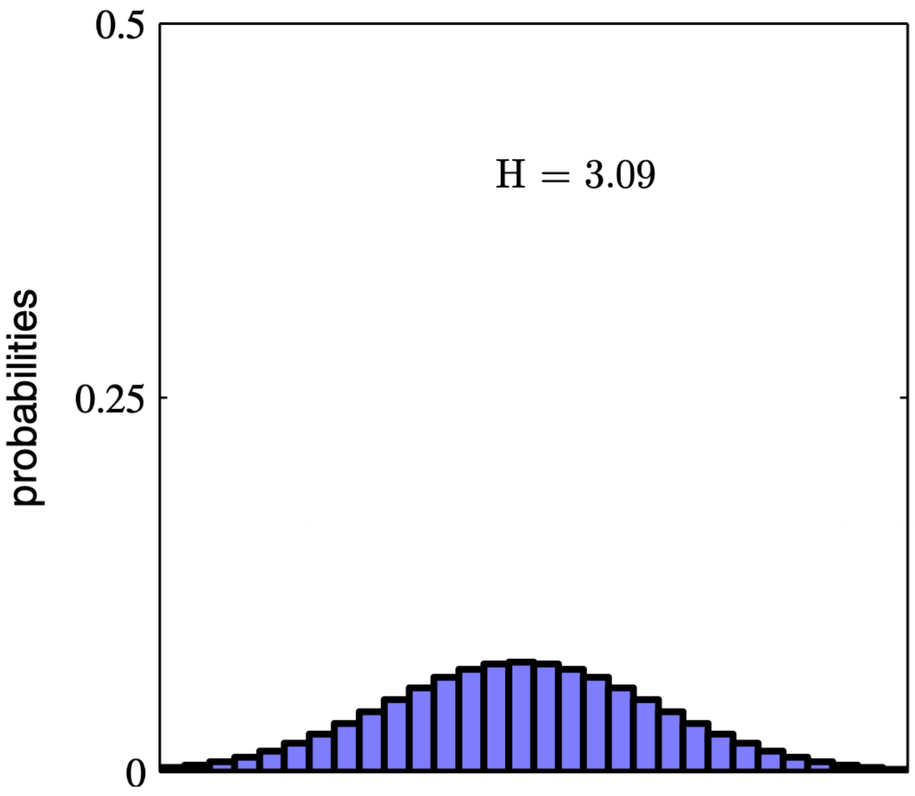
$$H(Y) = \sum_i p(x_i) \ \ln p(x_i)$$

$$H(Y) = -\,p\ (\log_2 p) - (1-p)\ (\log_2(1-p))$$



Entropy of Two Probability Distribution Examples





Which distribution (in general) have the largest entropy?

Sample Entropy

Consider i.i.d. dataset of $\{y_i, \boldsymbol{x}_i\}_{i=1}^n$, where $y_i \in \{0,1\}$ and $\boldsymbol{x}_i \in \mathbb{R}^p$.

Let $D = \{(y_i, \boldsymbol{x}_i) : i = 1, \ldots, n\}$.

Suppose $p(y_i = 1) = p$, with sample estimate of the parameter p as

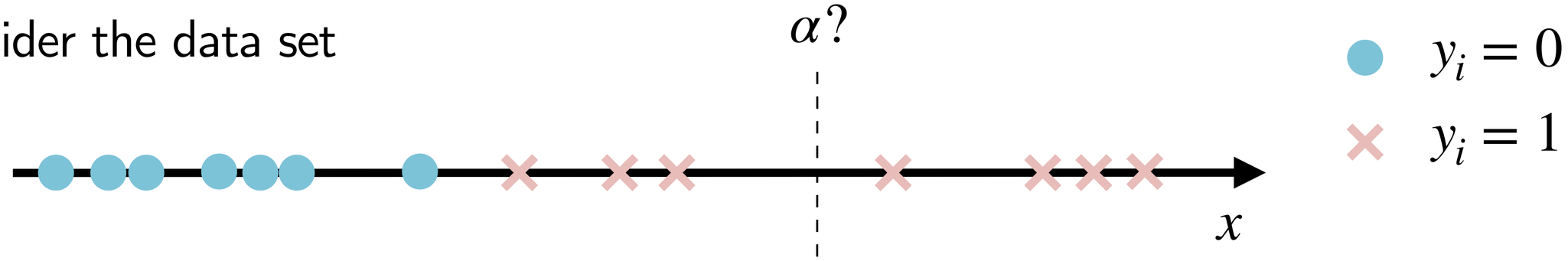
$$\hat{p} = \frac{\sum_{i=1}^n y_i}{n}$$

The **sample entropy** of the data, denoted by $H(D)$ is

$$H(D) = -\hat{p}\ \log_2 \hat{p}\ - (1 - \hat{p})\ \log_2(1 - \hat{p})$$

Example

Consider the data set



Suppose we split the data set into

$$D_\alpha^- = \{(y_i, x_i) : x_i \leq \alpha\} \qquad D_\alpha^+ = \{(y_i, x_i) : x_i > \alpha\}$$

The entropy of the data after splitting is

$$H(D_\alpha) = H(D_\alpha^+)\,\frac{|D_\alpha^+|}{|D|} + H(D_\alpha^-)\,\frac{|D_\alpha^-|}{|D|}$$

Where should we put α to minimize the entropy?

Decision Tree Classification Algorithm

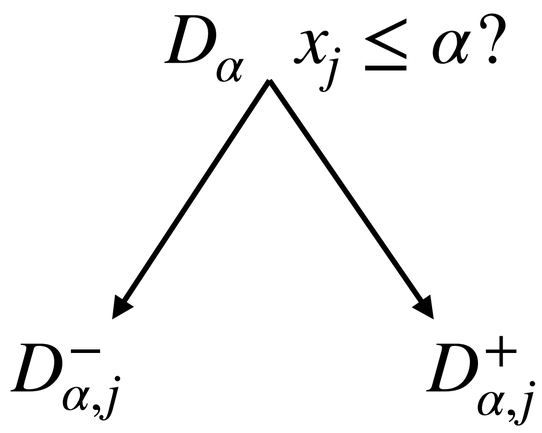
Consider i.i.d. dataset of $\{y_i, \boldsymbol{x}_i\}_{i=1}^n$, where $y_i \in \{0,1\}$ and $\boldsymbol{x}_i \in \mathbb{R}^p$, and define $D = \{(y_i, \boldsymbol{x}_i) : i = 1, \ldots, n\}$.

We define

for coordinate j

$$D^-_{\alpha,j} = \{(y_i, \boldsymbol{x}_i) \mid \boldsymbol{x}_{i,j} \leq \alpha\} \qquad D^+_{\alpha,j} = \{(y_i, \boldsymbol{x}_i) \mid \boldsymbol{x}_{i,j} > \alpha\}$$

We think of D as the root of a binary tree, and $D^+_{\alpha,j}$ and $D^-_{\alpha,j}$ as the two child nodes that result from splitting along the jth feature.



Was the Decision Partitioning Useful?

We quantify the "usefulness" of the branch/partition using information gain.

Information Gain

We define the information gain from the splitting of D at α along the jth feature dimension as

$$I_{\alpha,j} = \underbrace{H(D)}_{\text{Uncertainty/entropy before splitting.}} - \underbrace{[H(D_\alpha^+)\frac{|D_{\alpha,j}^+|}{|D|} + H(D_{\alpha,j}^-)\frac{|D_{\alpha,j}^-|}{|D|}]}_{\text{Uncertainty/entropy after splitting.}}$$

*Uncertainty/entropy **before** splitting.*

*Uncertainty/entropy **after** splitting.*

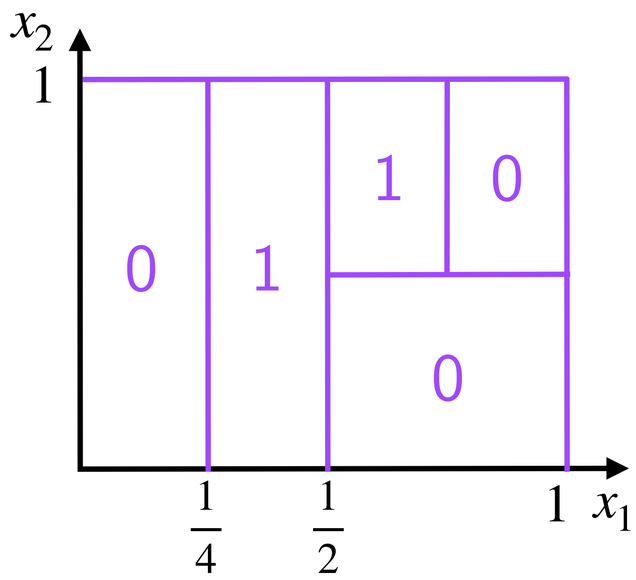
Basic Decision Tree Algorithm for Classification

1. Start with data set D.

2. **Recursively** split the data set along the feature and value (α) that result in **maximum information gain**.

3. Stop when maximum information gain is less than a specified threshold ΔH or depth.

Decision Tree Example



$$(x_1, x_2) \sim U([0,1] \times [0,1])$$

- Initially, $p(y = 1) = \frac{3}{8}$

$$H(D) = -\frac{3}{8}log_2(\frac{3}{8}) - \frac{5}{8}log_2(\frac{5}{8}) \quad = 0.954$$

Scenario A: split at $\alpha = \frac{1}{4}$, along x_1

$$I_{\frac{1}{4},1} = H(D) - \frac{1}{4}0 - \frac{3}{4}(-\frac{1}{2}log_2(\frac{1}{2}) - \frac{1}{2}log_2(\frac{1}{2})) = 0.204$$

Scenario B: split at $\alpha = \frac{1}{2}$, along x_2

$$I_{\frac{1}{2},2} = H(D) - \frac{1}{2}1 - \frac{1}{2}(-\frac{1}{4}log_2(\frac{1}{4}) - \frac{3}{4}log_2(\frac{3}{4})) = 0.0488$$

Which splitting criteria is better?

Takeaways

- ***Decision Tree***

  — *Partitioning your feature space sequentially such that the entropy and uncertainty is decreased*

  — *Maximize information gain at every branch of the tree*

  — *Read more on* ***Information Theory*** *and Shannon Coding, and "A Mathematical Theory of Communication", by Claude Shannon (this last bullet point is only for your information).*